ETH

D PHYS

87 - Acoustic waves and measurement of the sound velocity

Leopold Goedhart

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Abstract

In this experiment waves and interference phenomena were investigated using acoustic waves, which have a longer wavelength than electromagnetic waves and are therefore more suitable for an experiment performed at home due to COVID-19. Over the course of this experiment the properties of acoustic waves are analyzed, sound is detected with a smartphone and then analyzed regarding its frequency spectrum and resonance behavior. Making use of a special technique, the velocity of sound is determined only with a smartphone and a cardboard tube and finally the phenomenon of a standing wave is discussed. The experiment itself consists of three main parts. First, the resonance frequencies of the cardboard tube, with both ends open and with one open and one closed, are measured. Second follows the determination of the speed of sound and third an acoustic standing wave is created inside the tube in order to analyze the sound pressure level (SPL) inside.

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1 Introduction

In this experiment the behavior of sound waves in a tube is studied. Using a very simple, home-made set up including a cardboard tube, a smartphone with a microphone and the software "PhyPhox" the physical background of wave phenomena are explored. According to Wikipedia, a wave is a propagating dynamic disturbance (change from equilibrium) of one or more quantities, or in our case, air-molecules. There are two types of waves, longitudinal waves and transverse waves. A sound wave propagates parallel to the direction of the molecule displacement and is therefore a longitudinal wave. These sound waves can be detected by a microphone using the following principle. The so-called diaphragm in the microphone oscillates according to the approaching sound waves and passes this continuous signal on to other parts in the microphone. These other parts transform the diaphragm-signal into an electronic signal, which can be interpreted by an observer.

In the first part of this experiment, resonance phenomena are investigated using a tube open at both ends and a tube open only on one end. Exciting the resonance frequencies of air molecules in side the tube at one end allows for the detection of the resonance frequencies at the other end of the tube. In order to obtain the data in frequency-space, a Fourier transform is performed. The fundamental frequency is then extracted by fitting the lorentian function given by equation (1) to the data:

$$l(f) = y_0 + \frac{2A}{\pi}\frac{w}{4(f-f_0)^2 + w^2}, \tag{1}$$

where y_0 is the offset, f_0 is the center position, w is the width of the peak and A is the area under the peak. This procedure is then repeated for the same tube but with only one open end and one closed end.

In the second part of the experiment the velocity of sound is determined. The fundamental frequency f_0 and the higher orders of a standing wave in a tube, where both ends are either open or closed is given by equations (2) and (3):

$$f_0 = \frac{c_s}{2(L+2a)} \tag{2}$$

$$f_k = (k+1)f_0; k = 0, 1, 2, ..., \tag{3}$$

where c_s is the velocity of sound, L is the length of the tube and $a = 0.6133R$ ($R \equiv$ radius). The fundamental frequency and higher orders of a tube open on only one end are given by equations (4) and (5):

$$f_0 = \frac{c_s}{4(L+a)} \tag{4}$$

$$f_k = (2k+1)f_0; k = 0, 1, 2, ... \tag{5}$$

The given equations yield the velocity of sound, which can therefore be obtained by determining the first harmonic and the harmonics of higher orders for a more precise result. In order to determine these frequencies, a multi-lorentian function (see equation (6)) is fitted to the data obtained from the measurements.

$$m(f) = y_0 + b_0 \exp\frac{f}{b} + \frac{2A_0}{\pi}\frac{w_0}{4(f-f_0)^2 + w_0^2} + \frac{2A_1}{\pi}\frac{w_1}{4(f-f_1)^2 + w_1^2} + ..., \tag{6}$$

using the theoretical values presented obtained from the equations presented above. This is then followed by a linear fit of those maxima. The velocity of sound is then given by the slope of that linear fit.

In the third part of the experiment the sound waves for the resonance condition inside the tube and the standing wave between a frequency source and a wall are visualized. Using the audio amplitude function of PhyPhox, the sound pressure level (SPL) is measured as a function of time. By moving the microphone inside the tube, the SPL can be obtained as a function of the position in the tube. In the case of a tube against a wall, the SPL is measured as a function of the distance to the wall. This qualitative measurement can also be performed more quantitatively, by determining the position of the minima in the SPL between the wall and the frequency source.

2 Experiment

In the first part of the experiment, the resonance frequencies of the air inside a cardboard tube with a length L of 25.5cm and a radius R of 4cm were excited by blowing over the edge of one end of the tube. The microphone of the smartphone was distanced $0.6133 \times R$ away from the other end of the tube.

In order to perform the measurements for the tube with only one open end, the other end was sealed with cling film. The tube was then excited by rubbing the end of the closed end. The frequency spectrum was then measured with the PhyPhox application.

In the second part of the experiment, the data obtained from the first part was used in order to calculate the velocity of sound by plotting the resonance frequencies as a function of their corresponding order. Fitting a linear function to the data obtained, one can calculate the velocity of sound by using the slope m of the linear fit, corresponding to $m \equiv \frac{f_k}{k+1}$, where f_k is the resonance frequency of order k. Using equations (2), (3), (4) and (5) the velocity of sound can then be determined.

In the third part of the experiment, a headset was placed inside the tube such that one could slide it from one end to the other. Then a tone generator application was used in order to generate a single frequency at the second harmonic resonance frequency of the tube. Using the "audio amplitude" function of PhyPhox, the SPL was measured as a function of time, while moving the headset with a constant velocity. This measurement was repeated for the fundamental frequency and the higher harmonics.

Positioning the single-frequency source 60cm away from the wall, the SPL as a function of the distance to the wall was measured by measuring the SPL as a function of time and moving the headset between the wall and the source with a constant velocity, simultaneously.

Finally, again, the audio amplitude function was used, while sliding the microphone away from the wall until a local minimum in the SPL was detected. The position of that minimum was obtained by shifting the microphone back and forth to find the exact position.

3 Part I - Resonance frequencies in a tube

3.1 Results & Data Analysis

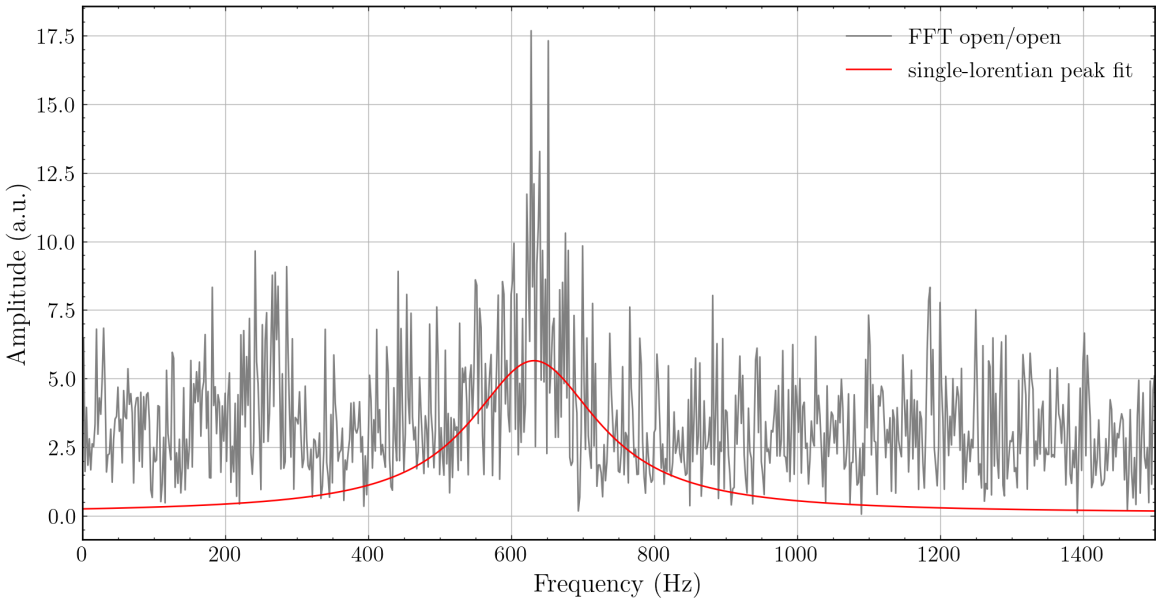

Figure 1: This figure shows the single-lorentian peak fit for the data obtained from the open/open tube. As expected, only the fundamental frequency was fitted. On top that, one can see that the noise is rather large, leading to a flat peak.

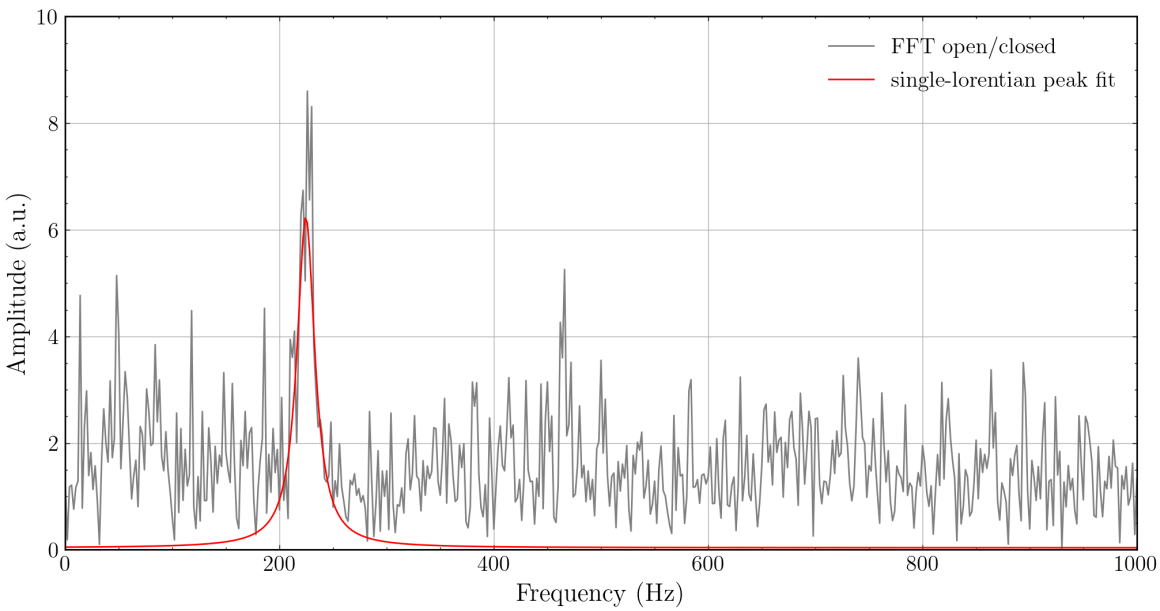

Figure 2: This figure shows the single-lorentian peak fit for the data obtained from the open/closed tube. Again, as expected, only the fundamental frequency was fitted, but here the noise is significantly smaller. Therefore the fitted peak is sharper.

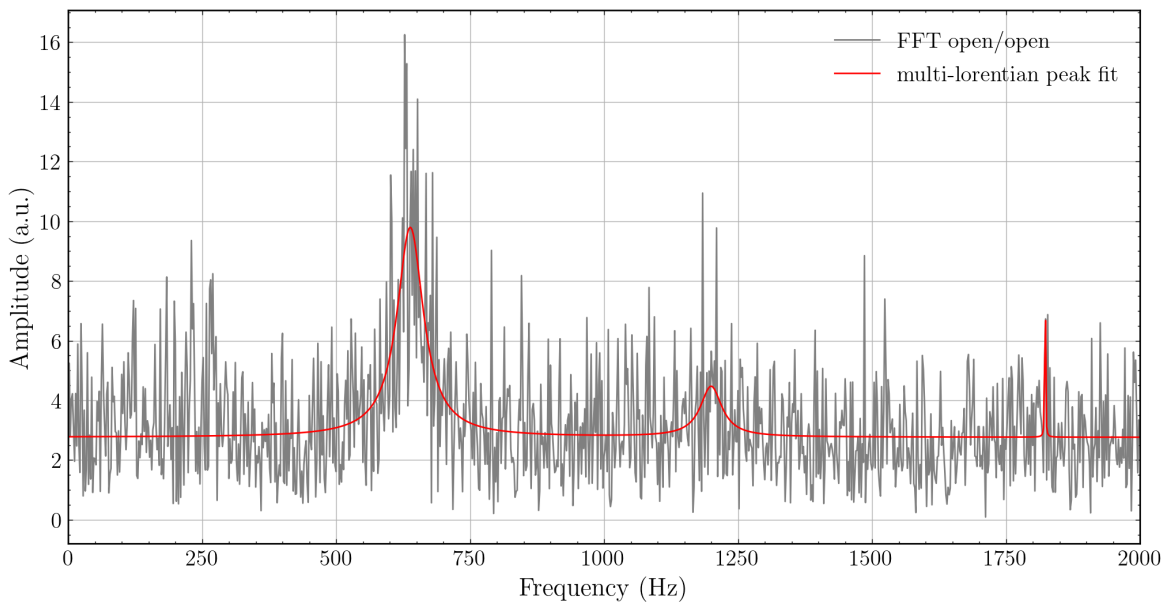

Figure 3: This figure shows the multi-lorentian peak fit for the data obtained from the open/open tube. It can be seen that there have been problems with the fit of the second and third peak. During the experiment, it was tried to resolve the problems by varying the initial fit conditions. Nevertheless, the fit did not work as expected. This is probably again explained by the large noise.

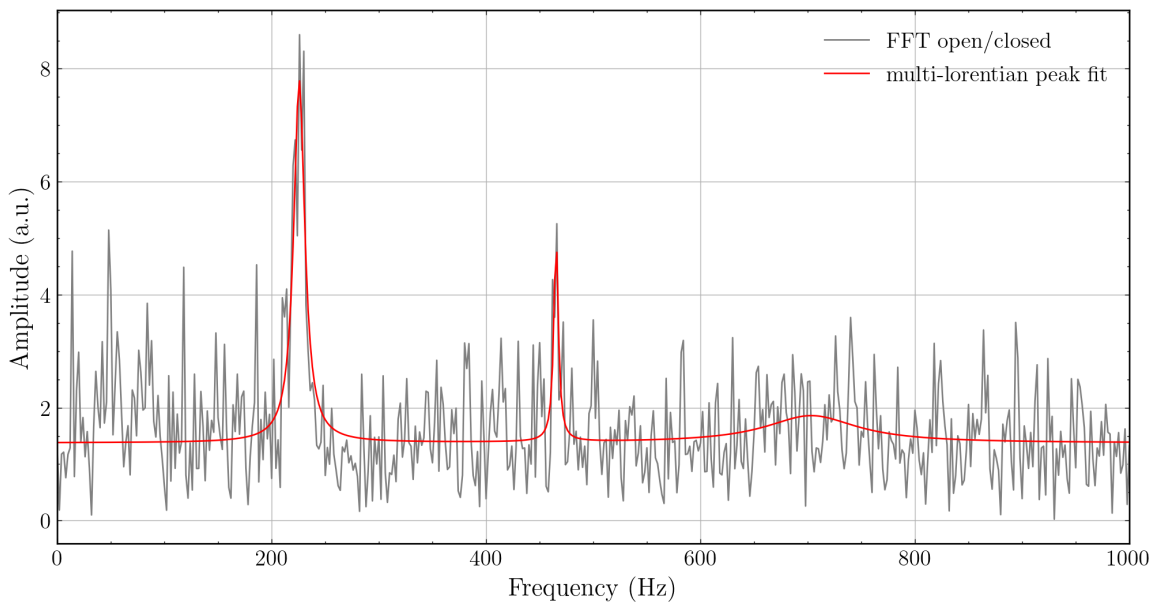

Figure 4: This figure shows the multi-lorentian peak fit for the data obtained from the open/closed tube. Here, it can be assumed that the fit worked fine, but the third harmony was not excited.

3.2 Discussion

The expected resonance frequencies obtained from equations (2), (3), (4), (5) can be seen in table (1). The frequencies obtained from the data obtained can be seen in table (2). They result from the multi-lorentian peak fit.

During the excitation process of the measurements I encountered severe difficulties. It was impossible to excite any resonance frequencies with the first setup. Only after consulting another team and trying to build a setup as close to their setup as possible, I was able to excite the resonance frequencies that can be seen in the figures presented above. The first setup included a plastic tube of length 25.5cm and a radius of 2.5cm. It was chosen with the idea that a plastic tube is more solid than a cardboard tube and therefore more cylindrical. Maybe the difficulties during the measurement can be explained by size of the first tube. Although the second setup performed better, as mentioned earlier, the resonance frequencies of higher order could not be excited enough, such that the multi-lorentian peak fit would consider them. This suggests that there was some disturbance in a component of the setup that was not changed. This could be too much surrounding noise or a fault in the microphone of the smartphone. While performing the measurements, I tried to minimize the surrounding sounds as mach as possible, but the life in a shared apartment can be disturbing.

	f_0 (Hz)	f_1 (Hz)	f_2 (Hz)
open/open	564 ± 2.9	1128 ± 6	1692 ± 9
open/closed	306.8 ± 1.3	920 ± 4	1534 ± 6

Table 1: This table shows the expected resonance frequencies. The uncertainties were calculated with the Gaussian Error Propagation with the uncertainties $\Delta L = \Delta R = 0.001$m.

	f_0 (Hz)	f_1 (Hz)	f_2 (Hz)
open/open	634.11 ± 3.15	1201.51 ± 5.51	1804.52 ± 3
open/closed	225.83 ± 0.55	465.08 ± 0.3	51.25 ± 3.05

Table 2: This table shows the measured resonance frequencies. The uncertainties result from the multi-lorentian peak fit. The third resonance frequency f_3 does not make sense in a physical context and suggests that the peak fit did not work as expected. This can be deduced from the imperfect data.

When comparing the expected frequencies with the measured ones, the tables suggest that for the open/open tube the measurement worked as expected, but for the open/closed tube, it did not. The results presented in tables (1) and (2) support the realization that the data fit was disturbed by noise and/or poor excitation. When comparing the values for the open/open tube, one can see that the measured frequencies are consistently above the expected value, which can be explained by the fact that the expected values are only calculations based on the assumption that the tube is perfectly cylindrical and that the excitation did not have noise, which was not the case. Therefore, the expected values may not represent the real resonance frequencies. As expected, the third resonance frequency of the open/closed tube that was measured does not make sense. We would have expected a frequency that is larger than the previous one. On top of that, the data fits yielded different results when performed several times. Therefore there is more uncertainty resulting from inequalities during that part of the calculation.

When comparing the frequencies for the two different types of tubes, one can see in the following figures (5) and (6) that for $k > 5$ the frequencies of the open/closed tube are larger than the

ones for the open/open tube, other than for $k \leq 5$. For the relation between the errors one can say that the difference between the two is getting larger for a larger k. On top of that, in both cases, the uncertainty is increasing for a larger k.

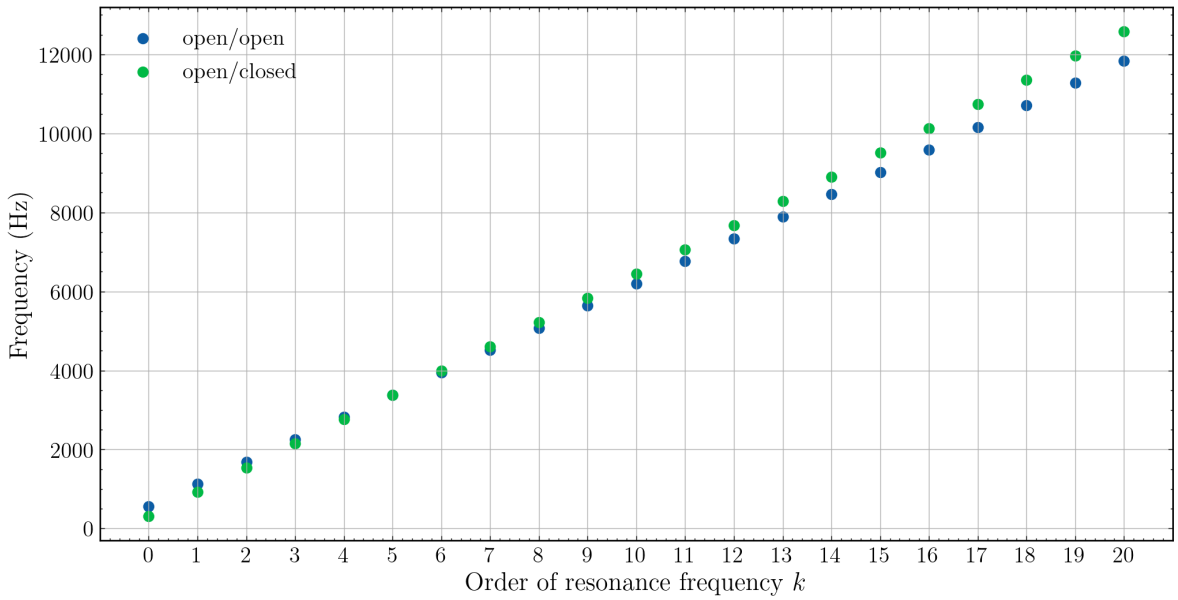

Figure 5: In this figure the theoretical resonance frequencies of the open/open and open/closed tube are compared.

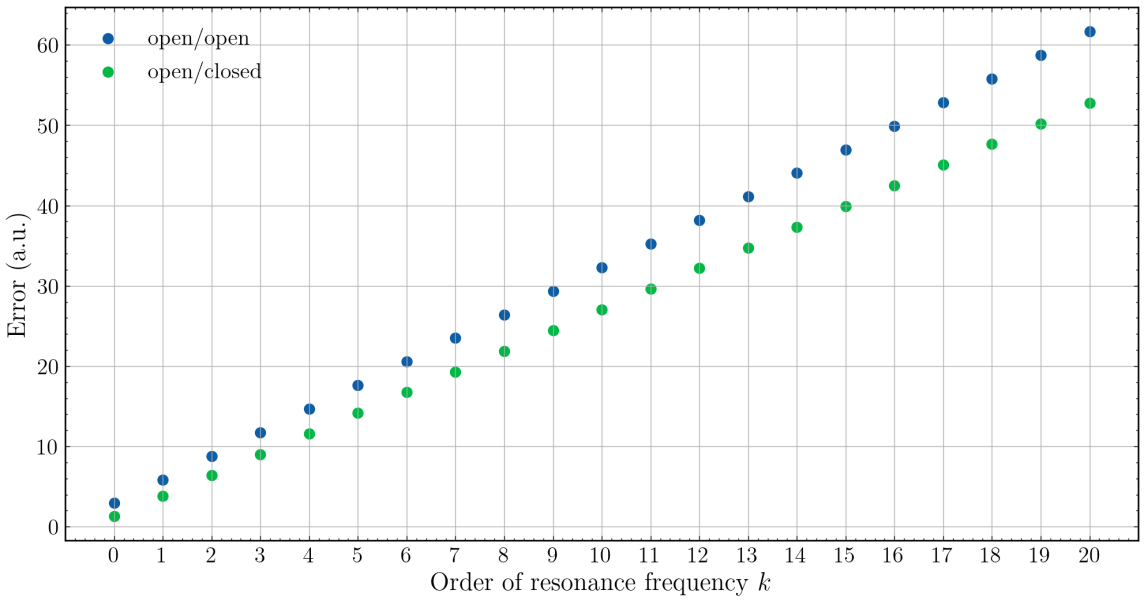

Figure 6: In this figure the uncertainties of the theoretical resonance frequencies of the open/open and open/closed tube are compared.

4 Part II - Measuring the speed of sound

4.1 Results & Data Analysis

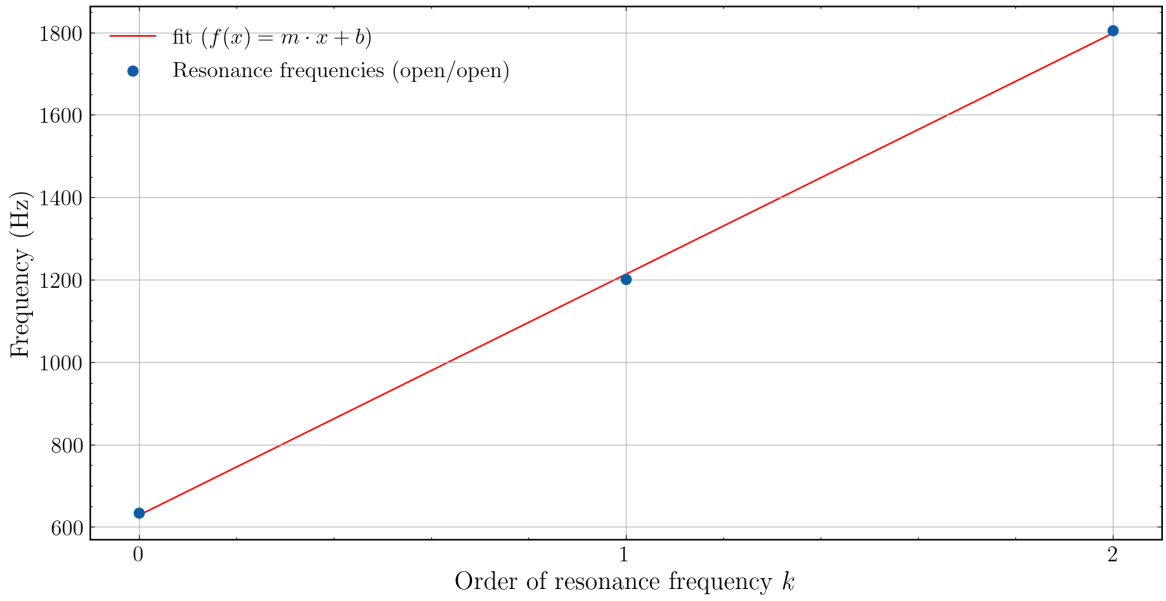

Figure 7: This figure shows the linear data fit mentioned in the Introduction (1) and yields the velocity of sound. The slope is given by $m = 585.21 \pm 10.28$.

The calculated velocity of sound is $v_{calculated} = 356 \pm 7\text{ms}^{-1}$, which is off by $\approx 3.8\% < 5\%$ compared to the literature value $v_{literature} = 343\text{ms}^{-1}$. The uncertainty was obtained with the Gaussian Error Propagation with $\Delta L = \Delta R = 0.001\text{m}$ and $\Delta m = 10.28$ from the multi-lorentian peak fit.

The velocity of sound obtained from the open/closed could not be calculated due to the fact that there were only two reasonable peaks fitted, as one can see in figure (4). For the calculation of the velocity of sound one would have to perform a "reindexation" of the data fit given by $k \mapsto 2k + 1$. As mentioned above, two peaks were not sufficient in order to perform this and therefore the data did not provide any reasonable result.

4.2 Discussion

The velocity of sound obtained from the open/open tube lies within a reasonable range (offset $< 5\%$) of the literature value and therefore speaks for a decent excitation of the resonance frequencies inside the tube. Unfortunately, this did not work for the open/closed tube, which can probably be explained by the fact that exciting the open/closed tube was much harder. The closing of one end yielded more free variables that increased the noise, as one can see in figure (4). When looking at the data obtained in the first part of the experiment one could have guessed that the results would be what they turned out to be, but after multiple attempts I gave up trying to get better data for the open/closed tube.

5 Part III - Characterizing an acoustic standing wave

5.1 Results & Data Analysis

First the acoustic wave inside the open/open tube was measured by mapping out the SPL as a function of the position in the tube, i.e. the distance to one opening. The fact that the SPL values are negative, can be explained by the microphone of the smartphone not being calibrated. The results are shown in the following figure (8).

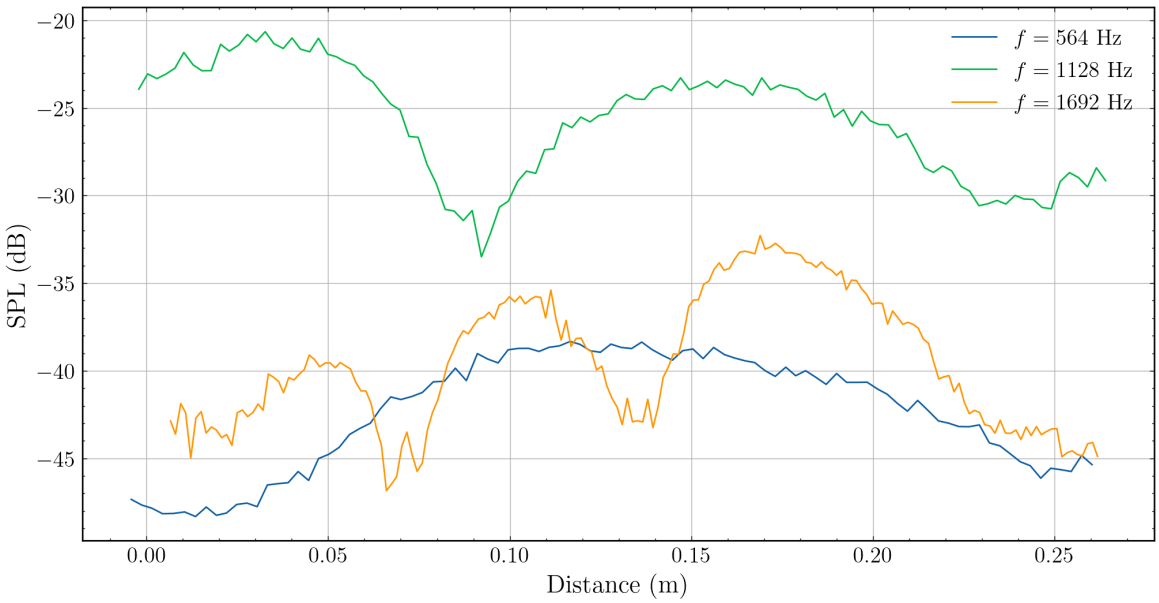

Figure 8: In this figure one can see the SPL as a function of the position inside the tube, i.e. the distance to one opening for three different resonance frequencies of the open/open tube.

Second, the acoustic standing wave between a frequency source and a wall should have been measured. As the results from the previous parts of the experiment show, the noise was rather large. Although I did everything possible to minimize the noise, the results are still affected by it. Due to this the results from this part of the experiment did not yield anything that could have been discussed or put into physical context. In order to substitute for this task, the measurement from before was repeated for the resonance frequencies of the open/closed tube. The results can be seen in figure (9).

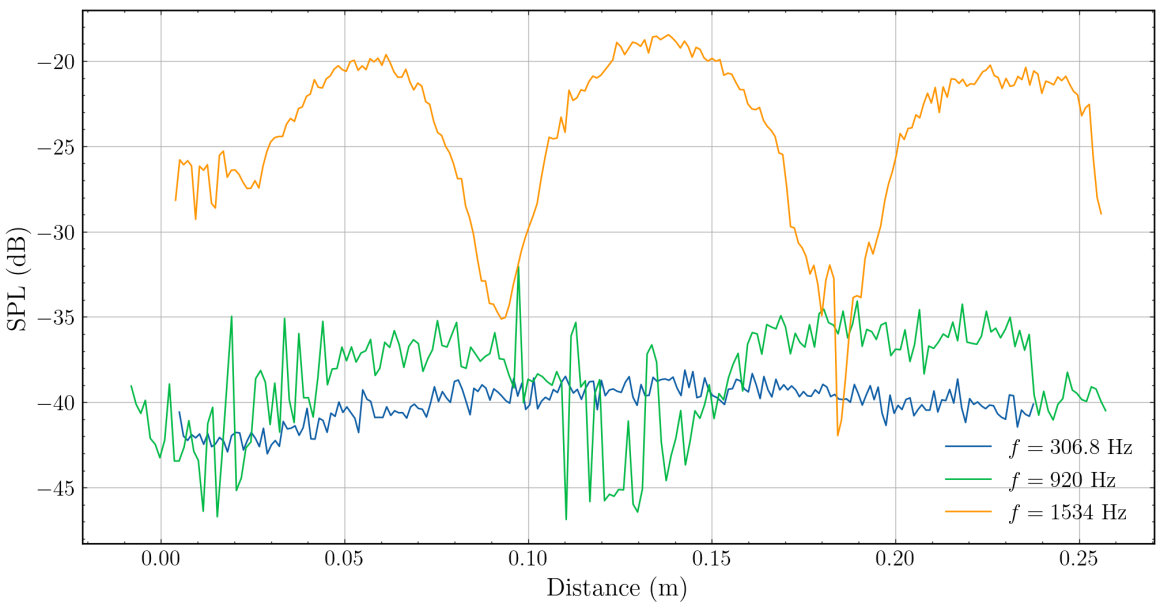

Figure 9: In this figure one can see the SPL as a function of the position inside the tube, i.e. the distance to one opening for three different resonance frequencies of the open/closed tube.

Here the tube was placed such that one end was closed by the wall and one end was open.

Third, the position of the minima in the SPL for different resonance frequencies of the standing wave were determined. In theory the minima of a standing waves occur at the nodes of the standing wave. This can also be seen in figure (10)[1]. As explained in the Experiment part (2) of this report, the minima were determined by approaching the open end of the tube with the microphone, while exciting the resonance frequency. When the SPL indicated a minimum by getting smaller, the microphone was moved back and forth. Initially the microphone was placed at distance 0cm from the open end. This was repeated twice, with the second and third resonance frequency. For the second resonance frequency ($k = 1$) of 920Hz, the position of the minimum was determined to be at a distance 14.5 ± 0.5cm and for the third resonance frequency ($k = 2$) of 1534Hz the first minimum was distanced 9.5 ± 0.5cm from the starting point and the second minimum was located at a distance of 20 ± 0.5cm.

[1]https://newt.phys.unsw.edu.au/jw/flutes.v.clarinets.html

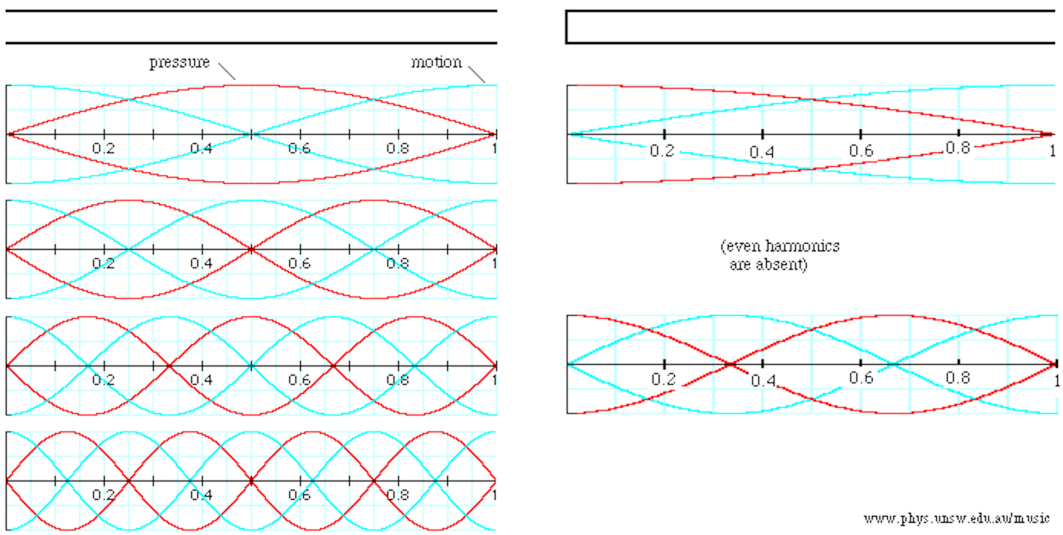

Figure 10: On the left one can see the amplitude of the variation in pressure in red and the amplitude of the variation in the flow of air in blue for an open/open tube for different harmonics. On the right one can see the same two properties of a open/closed tube for the uneven harmonics.

5.2 Discussion

In figure (8) one can see two minima located at the ends of the tube for $f = 564$Hz, three minima for $f = 1128$Hz and four minima for $f = 1692$Hz, as expected. On top of that, the minima in between the minima at the ends of the tube are approximately, symmetrically distributed between the two ends. This further supports the physics discussed before and the expected behavior regarding the total number of minima and the location of them, that can be seen in figure (10).

Theory suggests that a closed end of a tube is an antinode in the SPL. This can be perfectly seen in figure (9) for the excitation with the third resonance frequency, where there is a maximum at the closed end and a minimum at the open end and is also supported by figure (10) when looking at the right. There one can see that a maximum is located at the closed end and a minimum at the open end. On top of that, the other minima are again symmetrically distributed between both ends and can clearly be identified from the data.

The excitation with the first and second resonance frequency did not yield promising results. This may be explained by noise in that frequency range.

6 Conclusion

This experiment was originally designed so that it could be performed at home, only with household items. This was certainly not the case for me. I had enormous difficulties with the excitation of the resonance frequencies of the tube, although I followed the manual carefully and spent a lot of time into thinking about what to improve. Nevertheless, from this experiment I learned more than in any other experiment that good quality data is the key to meaningful results and analysis. On top of that, I learned that noise that is first thought of as negligible, will cause large problems after further calculations. Nevertheless, the data obtained from

the measurements yielded results that support the underlying physics and mostly work in accordance to the expectation.

All in all the experiment was very educational regarding fitting data with multiple variables, the propagation of uncertainty and noise and the improvement of a setup in the case of bad results.

7 Appendix

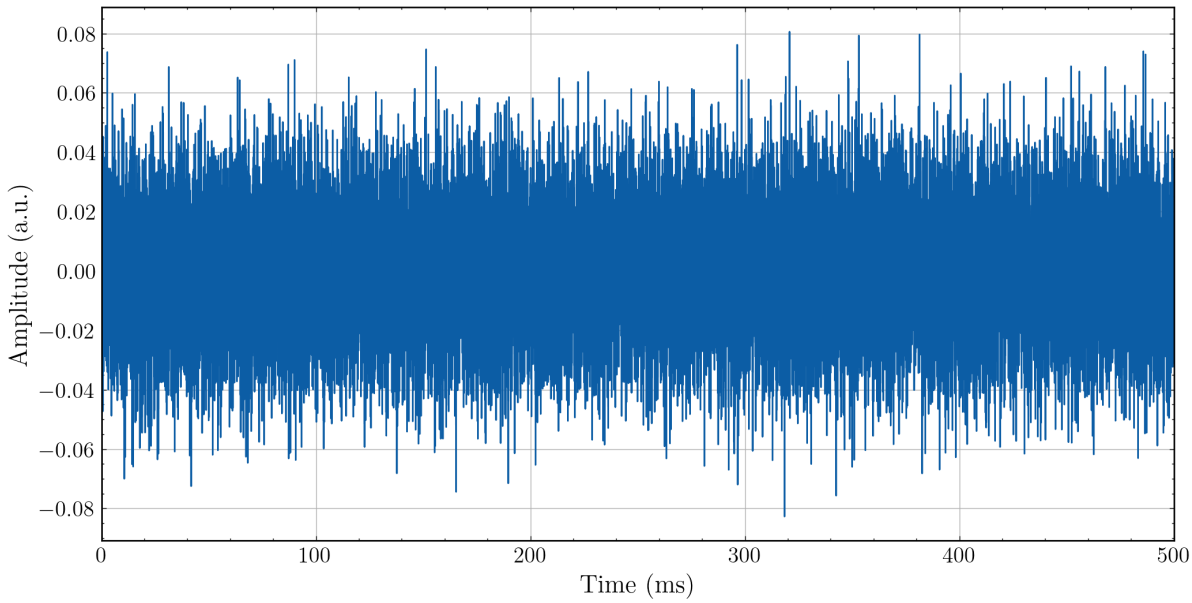

Figure 11: This figure shows the raw data of the measurements of the open/open tube.

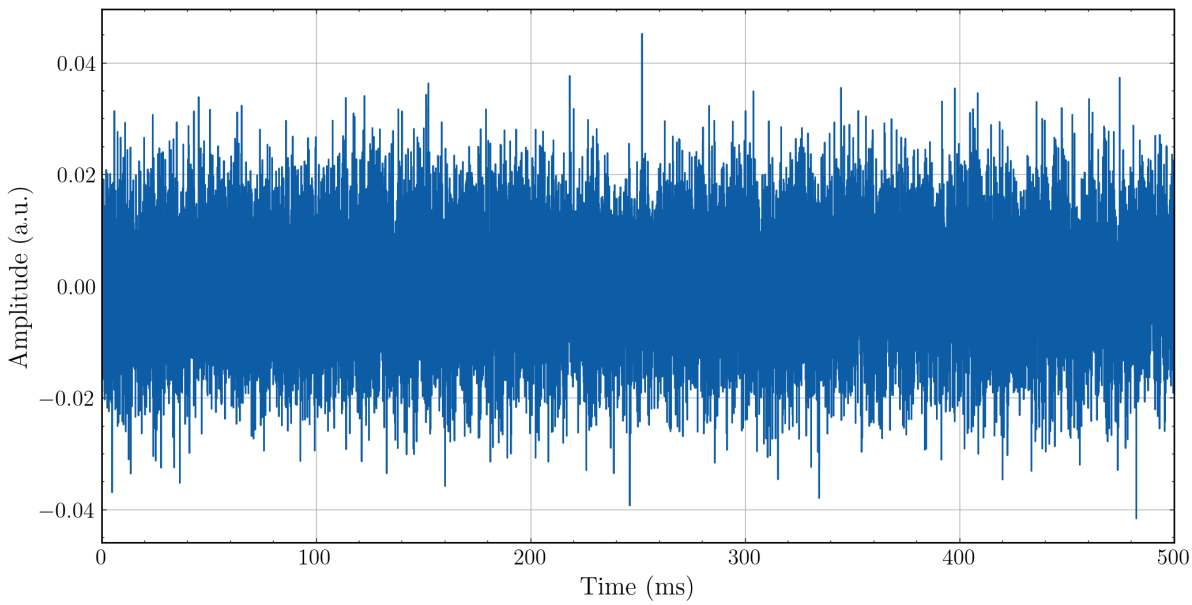

Figure 12: This figure shows the raw data of the measurements of the open/closed tube.